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Technical Manual

Ruggedized Field Telephone Set (RFT1001)

Part II, Technical Info – Maintenance

Doc Id: 1551/3-DPAR002014 (Rev. PA2)

(xxxxx-XXXXXX)

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Prepared and Published by:

Elcom Innovations Pvt. Ltd.
B-76, Phase 7, Industrial Area, Sector 74,
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Dec, 2013

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Ruggedized Field Telephone Set (RFT1001)

Part II, Technical Info – Maintenance

Doc Id: 1551_3-DPAR002014-PA1,

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INDIA

RECORD OF CHANGES

Change Notice No.	Authority	Incorporated by “Name and Rank” in Block letters	Initials with date
(1)	(2)	(3)	(4)

ASSOCIATED HANDBOOKS

Technical Literature for **FIELD TELEPHONE SET IFT-1001**
comprises of the followings:

- **USER HANDBOOK**
- **TECHNICAL MANUAL**

PART I TECHNICAL INFORMATION

Volume 1 Technical Description

Volume 2 Drawings

PART II MAINTENANCE

PART III OVERHAUL AND RECONDITIONING INSTRUCTIONS

PART IV MANUFACTURER'S PARTS LIST

Volume 1 Parts List

Volume 2 Illustrations

SAFETY WARNING

**The voltages employed in this Equipment are
sufficiently high to endanger human life.**

**POWER
MUST BE SWITCHED OFF
before servicing the equipment**

**&
GREAT CARE
taken when making internal adjustments etc.**

FOR FIRST AID IN CASE OF ELECTRIC SHOCK SEE OVERLEAF

FIRST AID IN CASE OF ELECTRIC SHOCK

FIG. 1.



FIG. 2.



FIG. 3.



FIG. 4.



1.	SWITCH OFF. If this is not possible, PROTECT YOURSELF with dry insulating material and pull the victim clear of the conductor. DO NOT TOUCH THE VICTIM WITH YOUR BARE HANDS until he is clear of the conductor, but DO NOT WASTE TIME.
2.	(a) Place the victim in the supine position. (b) Keep the air passages clear by turning the head to one side, opening the patient's mouth and clearing it of water, saline, mucus or blood, a lot of which might have accumulated in the back of the throat.(Figure 1).
3.	If the jaw is rigid try to force the mouth open by pressure on the gum behind the last molar tooth of the lower jaw. When the upper air passages are thus cleared, tilt the head backward and force the jaw forwards from the angles of the jaw in front of the ears. This would prevent mechanical obstructions to the upper air passages.(Figure 2-3)
4.	(a) Then hold the chin up and forward with one hand and pinch the nostrils of the victim with the other.(Figure 4) (b) Take a very deep breath and apply your mouth to that of the victim and blow into his mouth, until the chest of the victim moves up indicating filling of the lungs. (NEVER ALLOW THE CHIN TO SAG).(Figure 4)
5.	When the chest has moved up, withdraw your mouth and allow the chest to sink back. REPEAT this process every three to four seconds until the victim begins to breathe again or until he is taken over by a medical attendant. This method can be continued in an ambulance during transit of the patient from the site of accident to the nearest medical centre.
HAVE SOMEONE ELSE AND SEND FOR A DOCTOR. KEEP PATIENT WARM AND LOOSEN HIS CLOTHING. DO NOT GIVE LIQUIDS UNTIL THE PATIENT IS CONSCIOUS.	

DEMOLITION OF MATERIAL TO PREVENT ENEMY USE

(A) General

The demolition procedure outlined below will be used to prevent the enemy from using or salvaging this equipment. **DEMOLITION OF THE EQUIPMENT WILL BE DONE ONLY ON ORDER OF THE COMMANDER.**

(B) Methods of Destruction

Use any or all the following methods to destroy the equipment.

- (i) Smash - Smash the crystals, controls semiconductors, coil, switches, transformers and handset. Use sledges, hand axes, pickaxes, hammers crowbars and heavy tools.
- (ii) Cut - Cut the cords, headsets and wiring. Use axes, hands axes or hatchets.
- (iii) Burn - Burn the cords, resistors, capacitors, coils, wiring, technical manuals. Use gasoline, kerosene, oil, flamethrowers or incendiary.
- (iv) Bend - Bend the panels, cabinet and chassis.
- (v) Explosives - If explosives are necessary, use firearms, grenades or TNT.
- (vi) Disposal - Bury or scatter the destroyed parts in slit trenches, fox holes or throw them into streams.

CHANGE NOTICE

TITLE OF PUBLICATION	CODE NO. AND DATE	PREVIOUS CHANGE	CHANGE NOTICE NO. AND DATE

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(H) By Hand – correction	
(P) By pasting enclosed matter	
(A) Additional information – only to be filled in publication	

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1 Scope of the Document

1.1 General

- a) This Technical Manual Part II – Maintenance is prepared as an aid for user and trained personal to perform inspection and routine maintenance of the Field Telephones.
- Chapter 2 : Describes the necessary information about the routine tests to be conducted on the Field Telephone as a part of maintenance.
- Chapter 3 : Describes about the Test Equipment's and test jigs to be used during maintenance, setting up and alignment of the Field Telephone.
- Chapter 4 : Provides the detail about the Fault Diagnosis. It includes preliminary test, visual fault location and detailed checks for the same. A practical fault location schedule is provided including test result and minimum stage/system.
- Chapter 5 : Provides the information about typical test reading.
- b) Changes/additions which may have to be incorporated from time to time as supplement to this Technical Manual Part IV- Volumes 1 and 2 (to be called as Parts Lists) will be issued by **M/s. Elcom Innovations Pvt. Ltd** covering:
- Identification details of Sub-Assemblies which were not available at the time of preparations.
 - Allotment of DS/ Service Catalogue and Part nos. not included earlier.
 - Changes in the equipment due to modifications/ Updation.
- c) User should amend this book and record details of changes in the Record of Changes page.

1.2 Reference Document

[1] User Handbook	1553-DPAR002014
[2] Technical Manuals,	
PART I - Technical Information, Description (Vol 1 of 2)	1551/1-DPAR002014
Technical Information, Drawings (Vol 2 of 2)	1551/2-DPAR002014
PART II - Maintenance	1551/3-DPAR002014
PART III - Overhaul & Reconditioning Instructions	1551/4-DPAR002014
PART IV - Manufacturers's Parts List – Part Lists (Vol 1 of 2)	1551/5-DPAR002014
Manufacturers's Part List – Illustrations (Vol 1 of 2)	1551/6-DPAR002014

1.3 Abbreviations

AC	Alternating Current
ADC	Analog to Digital Converter
AH	Ampere Hours
AICS	Advanced Interoperability Communication System
AIS	Alarm Indication Signal
CAS	Channel Associated Signaling
CLI	Calling Line Identification

CPLD	Complex Programmable Logical Device
CPU	Central Processing Unit
DAC	Digital to Analog Converter
dB	Decibel
DC	Direct Current
DMA	Direct Memory Access
DTMF	Dual Tone Multiple Frequency
PCB	Printed Circuit Board
PCBA	Printed Circuit Board Assembly
UUT	Unit Under Test
WH	Wire Harness

1.4 Intended Audience

The intended audience of this document is the trained technical personal on Field Telephone Set RFT1001.

2 Maintenance Levels

Field Telephone RFT1001 is a low maintenance device which is designed to be operated, maintained and logistically supported according to a structure of five levels:

Level	Line	Performance	Performer	Placement
Level 1	1 st Line	Daily and periodical control	Operator	Field
	2 nd Line	Locating and replacing failing main unit.	Operator	Field
Level 2	3 rd Line	Testing and replacement of sub-units	Technicians	Field workshop
Level 3	4 th and 5 th Line	Component level repairing	Trained Engineers	Base Workshop

Table 1: Maintenance Levels

2.1 Level 1 Maintenance

The 1st and 2nd line maintenance is performed by the operator. The 1st line is preventive maintenance while 2nd line maintenance includes troubleshooting down to the functional unit.

Level 1 maintenance is described in chapter 0. It is covered in the User Handbook of RFT1001 also. Refer 1553-DPAR002014.

2.2 Level 2 Maintenance

The 3rd line maintenance is performed by departmental workshops or mobile workshops. The 3rd line maintenance includes troubleshooting down to module / card level and changing of the faulty module / card. The technicians must also be qualified to perform 1st & 2nd line maintenance. The 3rd line maintenance is described in chapter 4.

The 3rd line maintenance does not involve any soldering. Subunits that can be replaced at 3rd line maintenance are:

- Electronic Cards
- Wire Harnesses
- Mechanical Sub-Assemblies

2.3 Level 3 Maintenance

The level 3 maintenance (4th and 5th line maintenance) is performed by the Base Workshop. It comprises repairing of various assemblies or electronic cards.

Due to the complexity involved in the repairing of the electronic cards of RFT1001 & the costs for necessary tools, test equipment and spare parts storage, most organizations choose to perform level 3 maintenance at the factory. The description of level 3 maintenance is not under the scope of this manual.

All three maintenance levels are shown in the Figure 1.

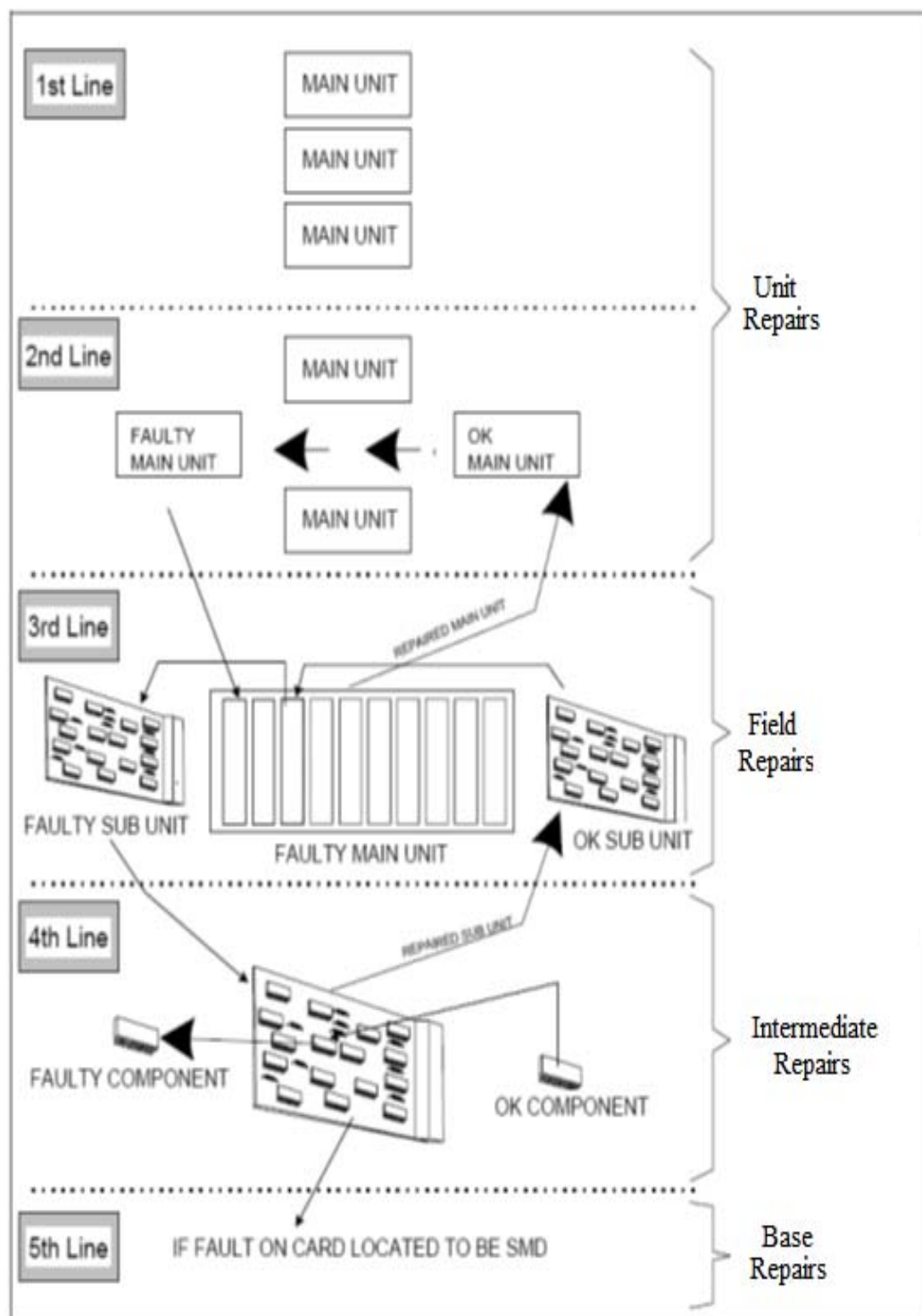


Figure 1: Maintenance Levels

3 Level 1 Maintenance

3.1 1st Line Maintenance

The 1st line maintenance shall be performed by the operator itself which includes preventive maintenance as explained in section **Error! Reference source not found.**

3.1.1 Tools, Test Equipment & Test Jigs, Required for 1st Line Maintenance

3.1.2 Daily Maintenance

- a) Clean the field telephone with a piece of clean dry cloth to remove dust.
- b) Clean the area between the **LINE** terminals with a hard nylon brush and check the line wires for proper gripping with the line terminals.
- c) Clean the area between the **EXTERNAL BATTERY** terminals with a hard nylon brush.
- d) Check the battery voltage by pressing **BAT MON** Key. If battery monitoring indicator 'P' lights up, replace all 3 batteries.
- e) Check the **PTT** by pressing it. A distinct click sound will be heard. Press **PTT** and blow into the microphone. Side tone will be heard in the receiver.
- f) Check the **MAGNETO CALL** button by pressing it. A distinct click sound will be heard.
- g) Connect a Magneto line. Press **MAGNETO CALL** button to call and talk with the other end's telephone by pressing PTT.
- h) Ask the other end's magneto telephone user to send a call signal and check if incoming call is detected by the field telephone.
- i) Connect an auto line. Lift the handset, hear the dial tone, dial a subscriber's number and talk.

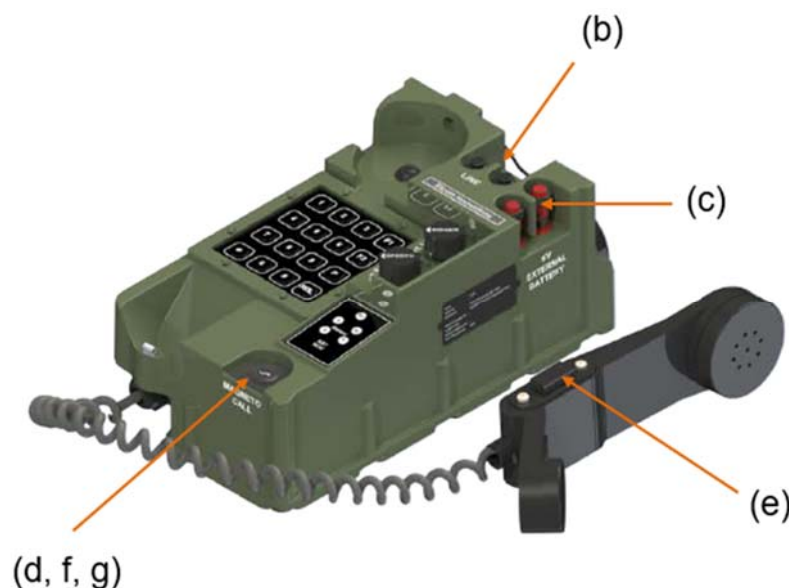


Figure 2: Daily Maintenance

3.1.3 Weekly Maintenance

- Repeat all the steps mentioned in the Daily maintenance routine. In addition, do the following:
- Open the battery box cap & take out all 3 dry batteries from the battery box.
- Clean the metal contacts of all 3 batteries with a hard nylon brush.
- After cleaning, put the batteries back in the battery box & tight the cap properly.



Figure 3: Weekly Maintenance

3.1.4 Monthly Maintenance

- Repeat all the steps mentioned in the daily maintenance routine and the weekly maintenance routing. In addition, do the following:
- Connect an AC voltmeter across the line terminals as shown in the Figure 24.
- Set the **SYSTEM** Switch in LB mode and press **MAGNETO CALL** button.
- AC voltage measured should be more than 90 volts.

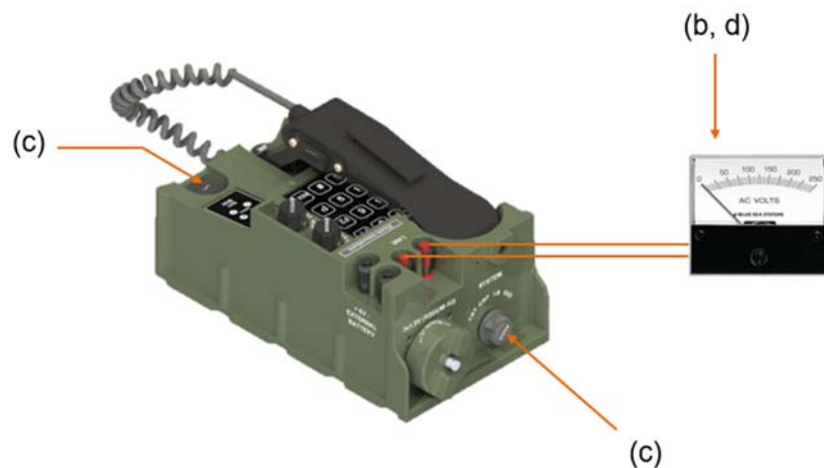


Figure 4: Monthly Maintenance

Note: If the RFT1001 is not in use for more than one month, remove all 3 batteries from the internal battery box to avoid damage caused by battery leakage.

3.2 2nd Line Maintenance

In 2nd line maintenance, operator can perform some tests manually to check the health of RFT1001. Before declaring the unit faulty, it is necessary to perform the preliminary test and visual fault location test on the unit. This chapter illustrates the tests which can be performed at field level.

3.2.1 Tools, Test Equipment & Test Jigs, Required for 2nd Line Maintenance

3.2.2 Fault Diagnosis - 2nd Line Maintenance

Step	Test Procedure	What happens	Conclusion
1	RING OUT Test (In LB Mode) – RFT1001 is not able to generate the ring in LB mode.		
a)	Connect a digital multimeter to UUT as shown in Figure 5.	-	-
a)	Keep the SYSTEM switch knob in LB position.	UUT is configured to work in LB mode.	-
b)	Keep the RINGER Switch to either Low or High position.	Ring volume is configured	-
c)	Press the MAGNETO Call button of the Field Telephone.	It will produce the Ring voltage across the Line terminals of the Field Telephone. It should be 90V at no load.	▪ If ring voltage is not 90V measured, replace the internal batteries & check again. If problem is still persists, UUT is faulty. ▪ For Level 1 maintenance, replace the faulty unit with good one & send the faulty unit to department workshop for Level 2 maintenance.
2	RING IN Test (In LB Mode) - RFT1001 is not able to detect the incoming ring in LB mode.		
a)	Prepare a test setup as shown in Figure 6.	-	-
b)	Keep the SYSTEM switch knob to position LB in both the Field Telephones.	Both the phones are configured to work in LB mode.	-
c)	Keep the RINGER Switch of Field Telephone B in Low or High position.	Ring volume of UUT is configured.	-
d)	Press the MAGNETO Call button of the Field Telephone A.	It will produce the ring at the Field Telephone B 'UUT' and the Ring Indicator LED (RV) starts flickering.	▪ If RV is not flickering, UUT is faulty. ▪ For Level 1 maintenance, replace the faulty unit with good one & send the faulty unit to department workshop for Level 2 maintenance.

Step	Test Procedure	What happens	Conclusion
e)	Add the 7500 Ohms resistance in series to field wires & again press the MAGNETO Call button of the Field Telephone A.	It will produce the ring at the telephone under test and the Ring Indicator LED (RV) starts flickering.	▪ If RV is not flickering, UUT is faulty. ▪ For Level 1 maintenance, replace the faulty unit with good one & send the faulty unit to department workshop for Level 2 maintenance.
3	Line Short Test		
a)	Prepare a test setup as shown in Figure 7Figure 6.	-	-
b)	Keep the SYSTEM switch in LB Mode.	-	-
c)	Short the LINE terminal using a shorting wire as shown in Figure 7.	Ring Indicator LED (RV) start flickering.	▪ If RV is not flickering, UUT is faulty. ▪ For Level 1 maintenance, replace the faulty unit with good one & send the faulty unit to department workshop for Level 2 maintenance.
d)	Remove the shortage from the LINE terminal.	Ring Indicator LED (RV) stop flickering.	▪ If RV is still flickering, UUT is faulty. ▪ For Level 1 maintenance, replace the faulty unit with good one & send the faulty unit to department workshop for Level 2 maintenance.
4	Speech & Incoming Ring Volume Control (In LB Mode)		
a)	Connect the two RFT1001 as shown in Figure 8.	-	-
b)	Keep the SYSTEM switch in LB mode in both the RFT1001 & keep the RINGER switch Low position.	-	▪
c)	Press the MAGNETO Call button on the Field Telephone A and keep it pressed.	Field Telephone B (UUT) rings at soft level.	▪
d)	While the Field Telephone B is ringing, rotate the RINGER Switch of the Field Telephone B to a) Mute Position b) High Position	a) Field Telephone B will not produce the ring. b) Ring volume of Field Telephone B will be increased.	▪ In both the cases the Ring Indicator (RV) will flicker. If it happens, UUT is Ok else faulty. ▪ For Level 1 maintenance, replace the faulty unit with good one & send the faulty unit to department workshop for Level 2 maintenance.

Step	Test Procedure	What happens	Conclusion
5	a) Pick the handset of Field Telephone B. b) Set RINGER knob to Med & High respectively. c) Press the PTT switch of Field Telephone B.	a) Field Telephone B 'UUT' will start hearing the voice of Field Telephone A. b) Incoming voice will be increased accordingly. c) Speech will be transmitted to Field Telephone A.	▪ If it happens, UUT is Ok else faulty. ▪ For Level 1 maintenance, replace the faulty unit with good one & send the faulty unit to department workshop for Level 2 maintenance.
5	Tone Dialing and Speech Test (In CBT Mode)		
a)	Connect the UUT to an Auto Exchange as shown in Figure 9.	-	-
b)	Set the SYSTEM switch knob of UUT to CBT mode for tone dialing.	-	-
c)	Lift the handset and listen the exchange dial tone.	'P/T' LED glows on UUT.	UUT is Ok as of now.
d)	Dial the required number from the dial keypad of the UUT.	a) Ring back tone will be heard if the called number is free. b) Busy tone will be heard if the called number is busy. d) If called number answer the call, UUT starts talking to called party.	UUT is Ok as of now.
e)	Once the conversation is over, replace the handset on its position	The P/T LED will be switched-off.	▪ If it happens, UUT is Ok else faulty. ▪ For Level 1 maintenance, replace the faulty unit with good one & send the faulty unit to department workshop for Level 2 maintenance.
6	Pulse Dialing and Speech Test (In CBP Mode)		
a)	Connect the UUT to an auto exchange as shown in Figure 10.	Setup is ready to perform the desired test.	-
b)	Set the SYSTEM switch knob of UUT to CBP mode for pulse dialing.	UUT is configured in CBP.	-
c)	Lift the handset & listen the exchange dial tone.	P/T' LED glows on UUT.	UUT is Ok as till now.

Step	Test Procedure	What happens	Conclusion
d)	Dial the required number from the dial keypad of the UUT. Since UUT is in CBP so number will be dialed in pulse.	a) While number is being dialed, P/T LED will flicker. b) Ring back tone will be heard if the called number is free. c) Busy tone will be heard if the called number is busy. If called number answer the call, UUT starts talking to called party.	▪ If UUT is able to talk to called party, UUT is ok else faulty. ▪ For Level 1 maintenance, replace the faulty unit with good one & send the faulty unit to department workshop for Level 2 maintenance.
e)	Once the conversation is over, replace the handset on its position	The P/T LED will be switched-Off.	▪ If UUT is able to talk to called party, UUT is ok else faulty. ▪ For Level 1 maintenance, replace the faulty unit with good one & send the faulty unit to department workshop for Level 2 maintenance.



Figure 5 : Test Setup for Ring Generator Test

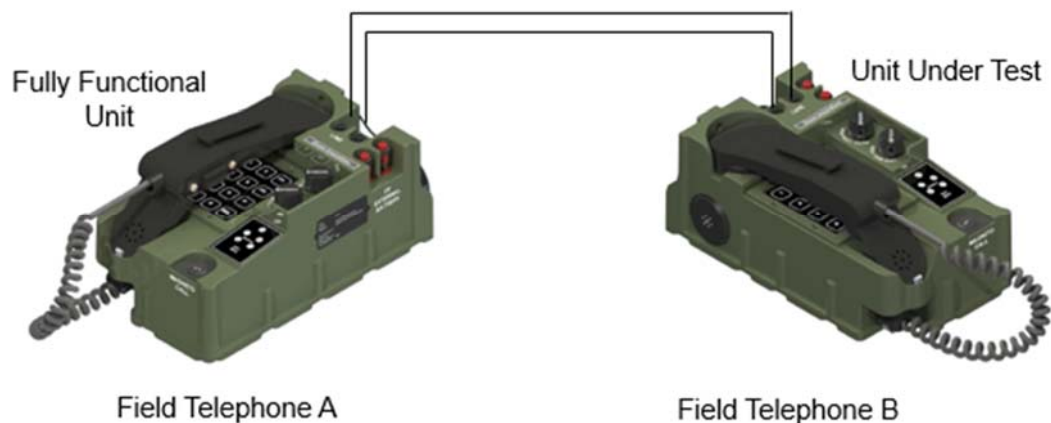


Figure 6 : Incoming Ring Test



Figure 7 : Line Short Test

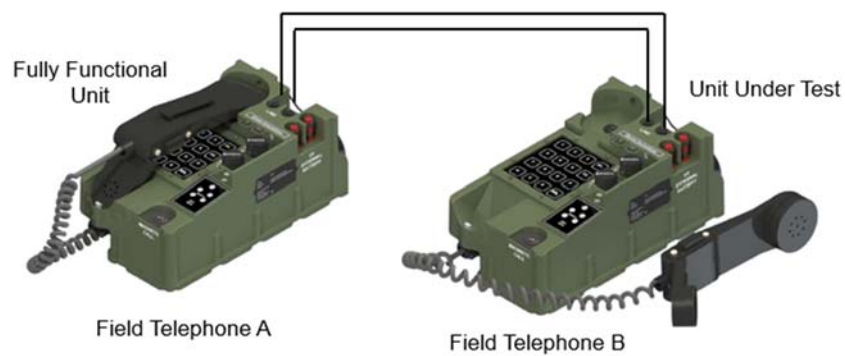


Figure 8 : Speech and Incoming Ring Volume Control

Figure 9 : Tone Dialing & Speech Test in CBT Mode



Figure 10 : Pulse Dialing and Speech Test (In CBP Mode)

4 Level 2 Maintenance

4.1 3rd Line Maintenance

Level 2 maintenance (3rd line maintenance) is performed at departmental workshops or mobile workshops. The 3rd line maintenance includes troubleshooting down to module / card level and changing of the faulty module / card. The technicians must also be qualified to perform 1stand 2nd line maintenance.

The 3rd line maintenance does not involve any soldering.

4.1.1 Test Equipment, Required for Level 2 Maintenance

Following is the list of the measuring instruments, tools and jigs required for the Level 2 Maintenance of the Field Telephone RFT1001.

4.1.1.1 List of Measuring Instruments

Description	Manufacturer Name and Model No.	Utilization	Required for
Variable DC Power Supply	Scientific	DC Power as external power source to operate RFT1001	Required for base workshop
Digital Multimeter	Rishabh, Rish Max 12	Use for measuring ▪ Ringer Voltage ▪ Ringer Frequency ▪ Resistance Check ▪ Capacitor Check ▪ DC Current Measurement	Required for monthly maintenance & base workshop
Audio Generator	APLAB	Injecting Tone input for checking system performance parameter.	Required for base workshop
Insulation Tester	Rishabh	Measures leakage between body and line terminal.	Required for base workshop
Telephone Analyzer	COSY/ Tayal TA- 2000	Use for measuring ▪ Pulse Count ▪ IDP ▪ Flash Period ▪ Incoming Ring Test	Required for base workshop

4.1.1.2 List of Tools

Description	Make & Model	Utilization	Required for
Screw Drover	Taparia / XXXXX	To screw/unscrew the Top assembly, PCBAs etc.	Daily, weekly & monthly maintenance.

Description	Make & Model	Utilization	Required for
Nose Plier	-	General purpose	Daily, weekly & monthly maintenance.
Medium size Nylon cleaning brush	-	General purpose	Daily, weekly & monthly maintenance.
Handset Socket tool	Elcom Innovations RFT1001 tool – Handset Socket	To dismantle the Handset Socket unit	
Line Terminal tool	Elcom Innovations RFT1001 tool – Line Terminal	To dismantle the Line terminals.	
System and Volume switch tool	Elcom Innovations RFT1001 tool – Switch	To dismantle the System and Volume switch.	

4.1.1.3 List of Electrical and Function Test Jigs

Description	Make & Model	Utilization
Test jig for the Main PCA Card	Elcom Innovations RFT1001 Test Jig - FT 001	Used for checking the speech circuit, buzzer circuit, supply circuit, battery monitor, status and ring indicator circuit.
Test jig for the Ringer PCA	Elcom Innovations RFT1001 Test Jig -FT 001	Used for verification of ringer circuit PCA for generating the ringer voltage.
Test jig for the Binding Post PCA	Elcom Innovations RFT1001 Test Jig -FT 001	Used for checking the functionality.
Test jig for the Hook switch PCA	Elcom Innovations RFT1001 Test Jig - FT 001	Used for checking the functionality.
Test jig for Handset assembly	Elcom Innovations RFT1001 Test Jig - FT 001	Used for checking the handset for overall operation, Microphone, Receiver and PTT.
Test Jig for Buzzer assembly	Elcom Innovations RFT1001 Test Jig - FT 001	Used for checking the operation of buzzer assembly

4.1.2 List of Various Assemblies

SA's Reference Id	SA Item No.	SA Description	SA Type
A1	EBA30058	Control Card	Electronic Card (PCBA)
A2	EBA30059	Ringer Card	Electronic Card (PCBA)
A4	EBA30062	System Switch Card	Electronic Card (PCBA)
S1	66085	Battery BOX Assembly	Mech SA (MSA)
S2	66086	Volume Switch Assembly	Mech SA (MSA)
S3	66087	Buzzer Assembly	Mech SA (MSA)
S4	66088	Buzzer Volume Assembly	Mech SA (MSA)
S5	66089	BAT MON Switch Assembly	Mech SA (MSA)
S6	66090	Ringer Micro Switch Assembly	Mech SA (MSA)
S7	66091	Keypad Assembly	Mech SA (MSA)
S8	66071	Line Cable Harness	Wire Harness Assembly (WHA)
S12	66076	Ringer Card Cable Harness	Wire Harness Assembly (WHA)
S15	66079	Hook Switch Harness	Wire Harness Assembly (WHA)
S16	66080	System Switch Harness	Wire Harness Assembly (WHA)
S18	66082	Ext Battery Cable Harness	Wire Harness Assembly (WHA)

4.1.3 Connection Diagram of Various Assemblies of RFT1001

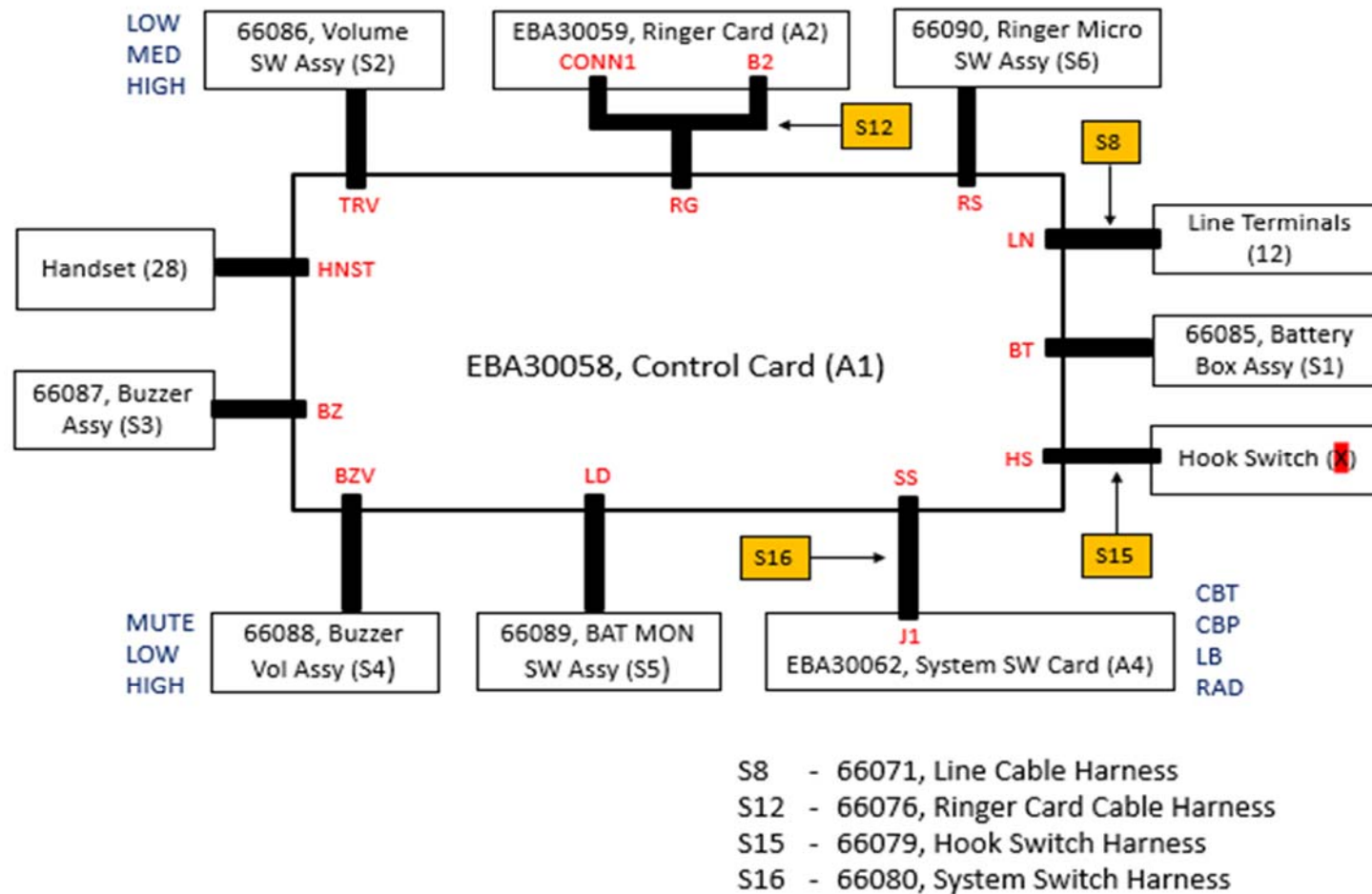


Figure 11: Connection Diagram of Various Assemblies

4.1.4 Important Guidelines for 3rd Line of Maintenance

- a) Always ensure that the handset on the faulty unit is working properly before start to follow the steps.
- b) It is mandatory that all harnesses remain connected so the signal path is established during the troubleshooting procedures.
- c) Chapter 4.1.5 shows trouble shooting steps for various problems. If step 1 fails to solve the problem, continue with step 2, and so on.
- d) It is difficult to do measurements and follow the signal path on connectors of carious cable harnesses. When relevant, alternative FRC connector reference points on other connectors or on components are used to do the measurements.
- e) Connectors on the card and cable harnesses are marked with a reference label for easy reassembly.
- f) This document is to do the troubleshooting & to diagnose the exact cause of fault. Once the faulty assembly is identified, document '1551/4-DPAR002014, Tech Manual-Part III, Overhaul & Reconditioning Instructions' shall be referred for dismantling & assembling of the faulty assembly. Even if UUT has to be sent for Level 3 Maintenance, assemble the removed assembly (if there is any) properly.

4.1.5 Fault Diagnosis - 3rd Line Maintenance

This chapter illustrates the fault diagnosis which has to be performed during 3rd line of maintenance. The fault which can affect the services of RFT1001 phones are:

- a) No Ring Out – Only in LB Mode
- b) No Ring In – CBT, CBP & LB Mode
- c) No Speech – In All Mode
- d) No Dialing Out – In CBT & CBP Mode

4.1.5.1 No Ring Out – Only in LB Mode

Consequence : No Outgoing Ring from UUT.
Test Setup : Connect UUT as shown in Figure 5.
Prerequisite : Ensure SYSTEM Switch is set to LB mode.
Outgoing Ring Path :

Figure 12: Outgoing Ring Path

Check Procedure* : After every step of troubleshooting, the MAGNETO CALL button of the UUT shall be pressed. If problem is resolved, connected multimeter will show 90V approx.

Troubleshooting Steps:

- Step 1 - Check 66085, Battery Box Assy (S1). There should not be any leakage. Check batteries also & check again. It should be approx 4.5 Volt.
- Step 2 - Replace below mentioned assemblies one by one in mentioned sequence. Check the UUT for the said problem after each part replacement.
- a) 66076, Ringer Card Cable Harness (S12)
 - b) EBA30059, Ringer Card (A2)
 - c) 66090, Ringer Micro SW Assy (S6)
 - d) 66080, System Switch Harness (S16)
 - e) EBA30062, System SW Card (A4)
 - f) EBA30058, Control Card (A1)
- Step 3 - If problem is resolved after replacing a specific assembly, faulty assembly is identified. Stop the troubleshooting further & dismantle/assemble the faulty identified assembly properly.

4.1.5.2 No Ring In – In CBT, CBP & LB Mode

- Consequence** : Incoming ring is not detected by UUT.
- Test Setup** : Take one fully functional RFT1001 as Ref phone & connect UUT as shown in Figure 6.
- Prerequisite** : Ensure **SYSTEM** Switch is set to LB mode.
Ensure Buzzer Vol Assy, 66088 (S4) is set to Low to High.
- Incoming Ring Path** :

Figure 13: Incoming Ring Path

- Check Procedure*** : After every step of troubleshooting, the MAGNETO CALL button of the reference telephone shall be pressed. If problem is resolved, UUT will start ringing.

Troubleshooting Steps:

- Step 1 - Verify 66071, Line Cable Harness (S8) for proper insertion. If any burnt sign is observed, replace WHA S8 & check* again.
- Step 2 - Replace below mentioned assemblies one by one in mentioned sequence. Check* the UUT for the said problem after each part replacement.
- g) 66087, Buzzer Assy (S3)
 - h) 66086, Volume SW Assy (S2)
 - i) 66080, SYSTEM Switch Harness
 - j) EBA30062, System SW Card (A4)
 - k) 66079, Hook Switch Assembly (S15)
 - l) EBA30058, Control Card (A1)
- Step 3 - If problem is resolved after replacing a specific assembly, faulty assembly is identified. Stop the troubleshooting further & dismantle/assemble the faulty identified assembly properly.
- Step 4 - Send the faulty assembly for Level 3 Maintenance.

4.1.6 Connectors and Harness Interconnection Matrix

The following chapters contains all details for connectors and harness cables on the various circuit cards inside the unit.

Description of reference / measurements points on the circuit cards to be used during fault finding procedures is also included and can be found in Chapter 4.1.6.5.

4.1.6.1 Overview

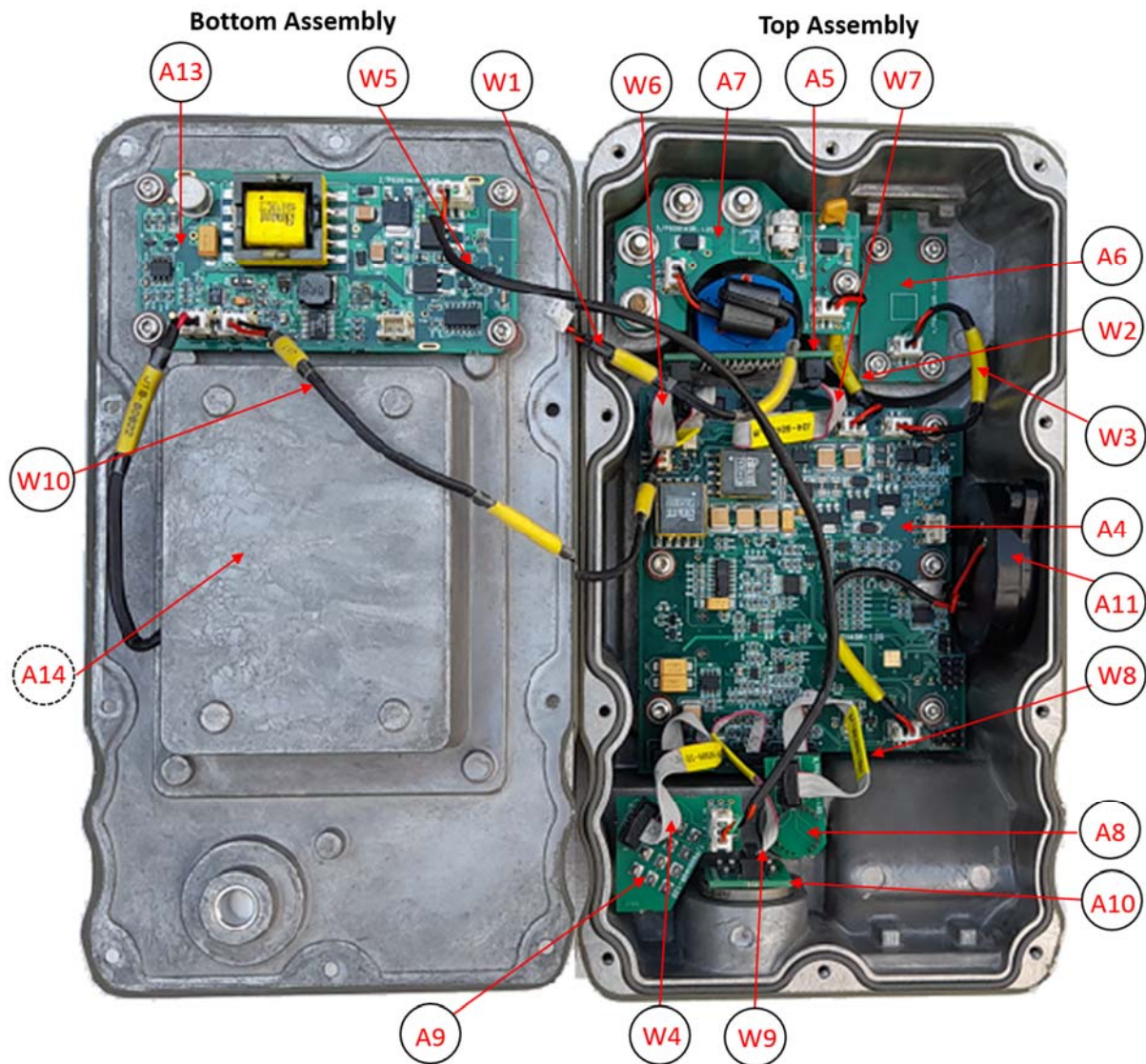


Figure 14: Overview of Open Unit

Legend to Figure 14:

- A4 - Main Card
- A5 - System Switch Card
- A6 - Reed Switch Card
- A7 - Binding Post Card
- A8 - Volume Switch Card
- A9 - Magneto Call Switch Card

A10	-	Handset Connector Card
A11	-	Buzzer Assembly
A13	-	Ringer Card
A14	-	Battery Box Assembly
W1	-	External Battery Harness
W2	-	Line Terminal Harness
W3	-	Reed Switch Harness
W4	-	Magneto Call Switch Harness-1
W5	-	Magneto Call Switch Harness-2
W6	-	System Switch Harness-1
W7	-	System Switch Harness-2
W8	-	Volume Switch Harness
W9	-	Handset Connector Harness
W10	-	Battery Harness

4.1.6.2 Harness Interconnection Matrix

Cable Ref./ Item ID	Cable From Card - Connector	Cable To Card - Connector
A11 (60611) Buzzer Assembly	Buzzer Assembly (A11)	Main Card (A4 -J3)
W2 (60624) Line Terminal Harness	Binding Post Card (A7 - J13)	Main Card (A4 – J1)
W3 (60625) Reed Switch Harness	Reed Switch Card (A6 - J14)	Main Card (A4 - J2)
W4 (60626) Magneto Call Switch Harness-1	Magneto Call Switch Card (A9 - J21)	Main Card (A4 - J6)
W6 (60627) System Switch Harness-1	System Switch Card (A5 – J23)	Main Card (A4 – J7)
W7 (60636) System Switch Harness-2	System Switch Card (A5 – J24)	Main Card (A4 – J9)
W8 (60628) Volume Switch Harness	Volume Switch Card (A8 - J20)	Main Card (A4 - J5)
W9 (60629) Handset Connector Harness	Handset Connector Card (A10 - J19)	Main Card (A4 - J4)
A14 (60622) Battery Box Assembly	Battery Box Assembly (A14)	Ringer Card (A13 - J16)
W1 (60833) External Battery Harness	Binding Post Card (A7 - J11)	Ringer Card (A13 - J12)
W5 (60635) Magneto Call Switch Harness-2	Magneto Call Switch Card (A9 – J22)	Ringer Card (A13 - J15)
W10 (60637) Battery Harness	Ringer Card (A13 - J17)	Main Card (A4 – J8)

Table 2: Harnesses, Connection Matrix

4.1.6.3 Top Assembly, Connector Locations

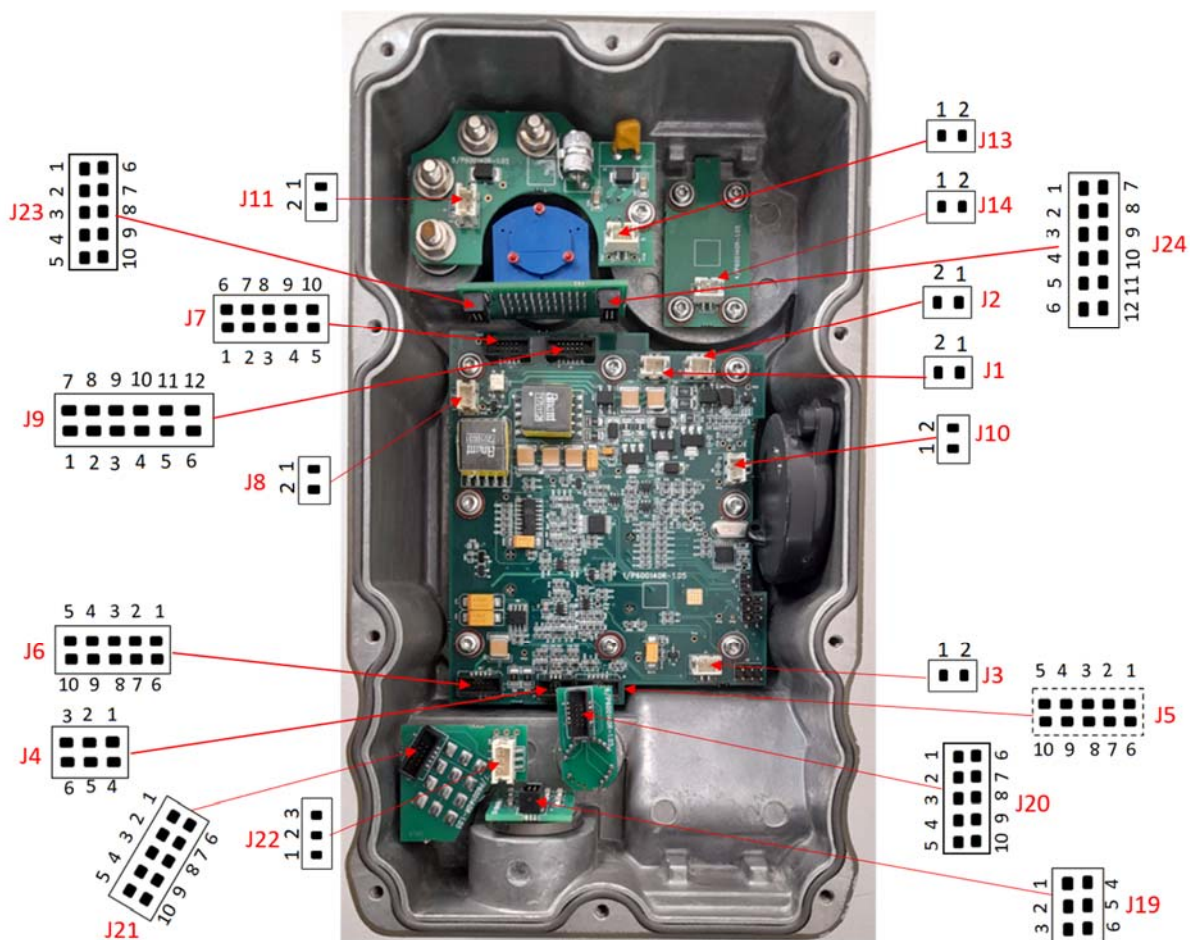


Figure 15: Top Assembly, Connector Locations

4.1.6.4 Bottom Assembly, Connector Locations



Figure 16: Bottom Assembly, Connector Locations

4.1.6.5 FRC Connector Pins and Components Measuring / Reference Points

4.1.6.5.1 Battery Box Assembly: Battery Pads

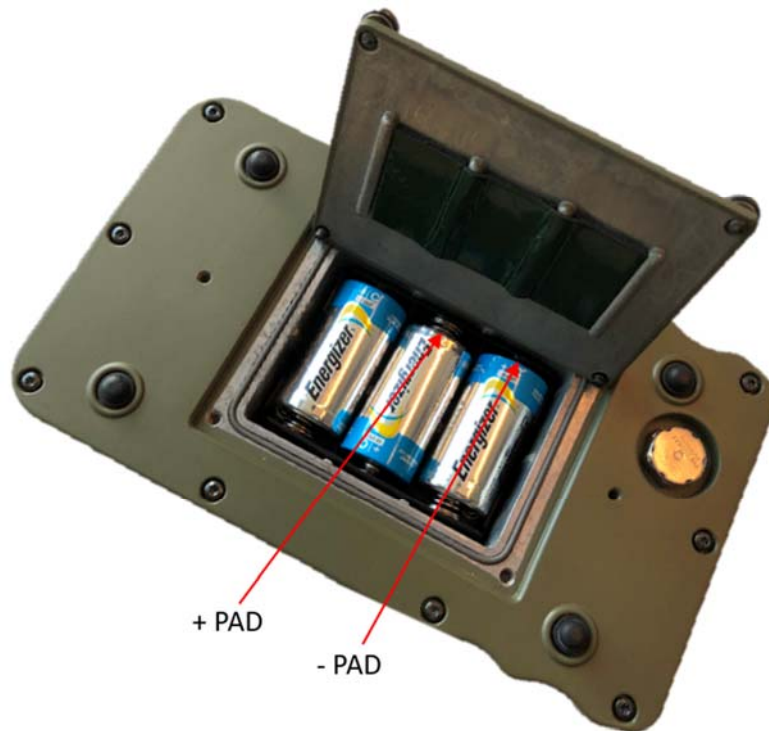


Figure 17: Battery Box Assembly, Battery Pads Measuring Points

4.1.6.5.2 Handset Connector Card: J25 Connector

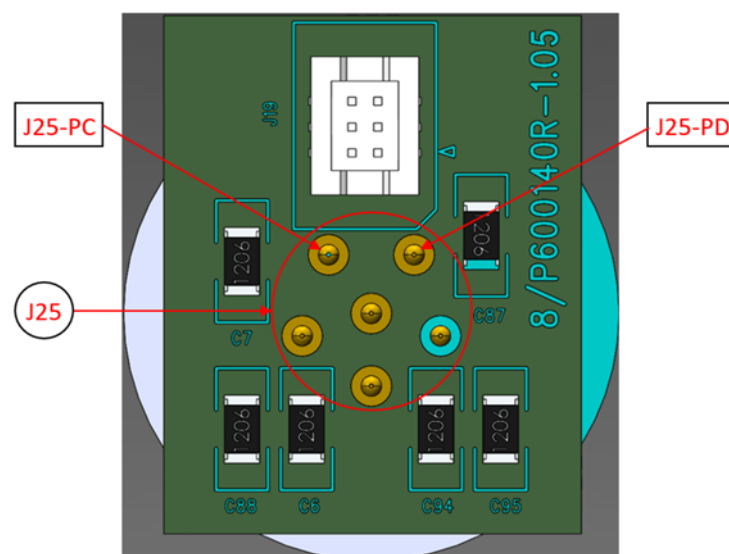


Figure 18: Handset Connector Card, J25 Connector Pins Measuring / Reference Points

4.1.6.5.3 Magneto Call Switch Card: SW5 Switch

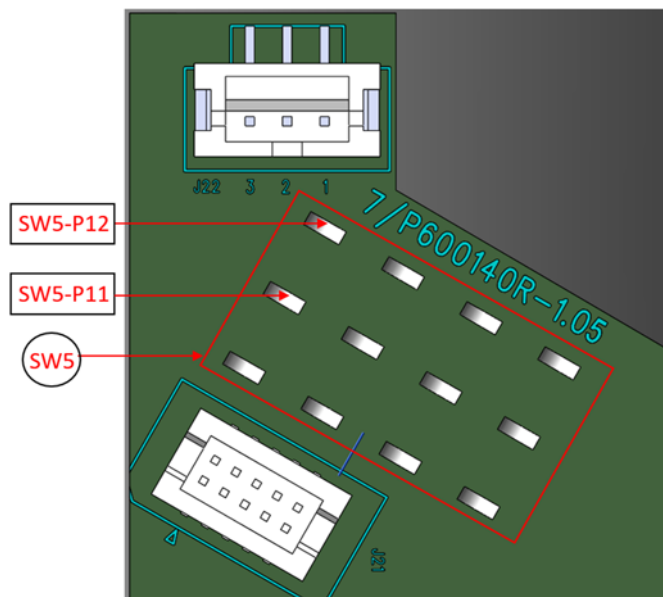


Figure 19: Magneto Call Switch Card, SW5 Switch Pins Measuring / Reference Points

4.1.6.5.4 Volume Switch Card: SW4 Switch

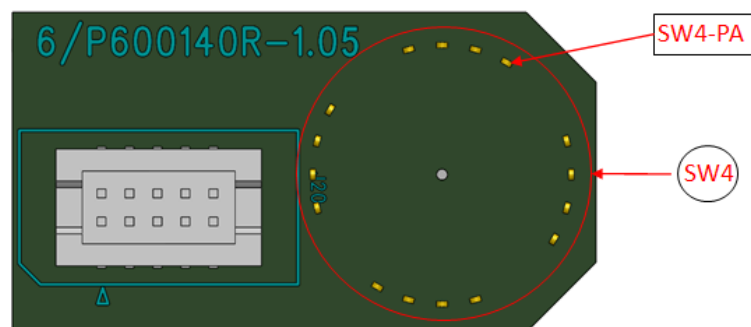


Figure 20: Volume Switch Card, SW4 Switch Pins Measuring / Reference Points

4.1.6.5.5 Main Card: Components

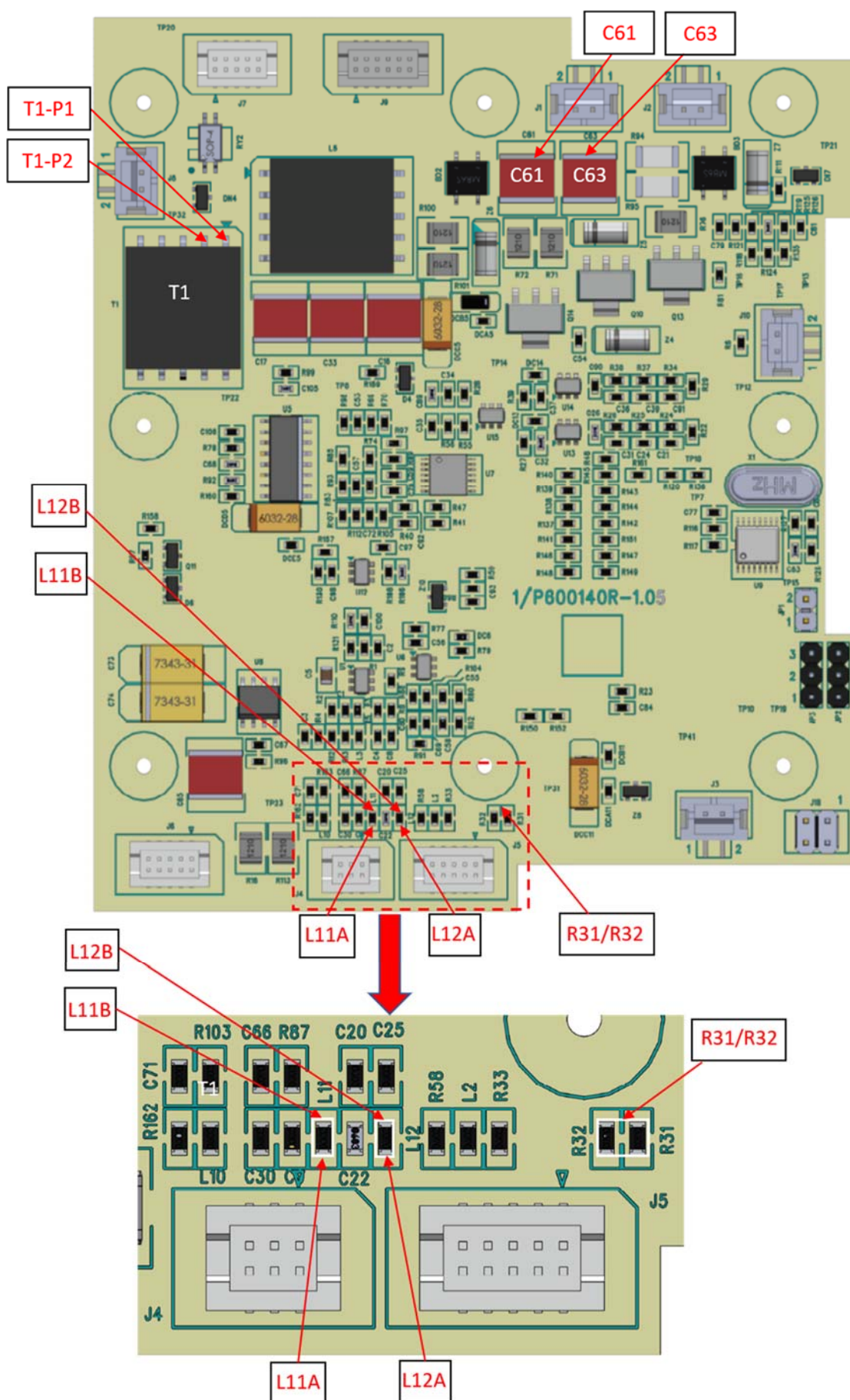


Figure 21: Main Card, Components Reference / Measuring Points

4.1.6.5.6 Measuring Points Versus Connector Pins in Signal Path

Measuring Points	Connector / Pin Reference	Card
C61	J6-P4	Main Card
C63	J7-P6	Main Card
L11A	J4-P1	Main Card
L11B	J7-P3	Main Card
L12A	J4-P2	Main Card
L12B	J7-P4	Main Card
R31/R32 (Common Terminal)	J5-P3	Main Card
T1-P1	--	Main Card
T1-P2	J9-P11	Main Card

Table 3: Components Versus FRC Connector Pins

Measuring Points	Connector / Pin Reference	Card
J1-P1	J6-P2, J7-P7	Main Card
J1-P2	J6-P6	Main Card
SW4-PA	J9-P12	Volume Switch Card - Main Card
J25-PC	J19-P1	Handset Connector Card
J25-PD	J19-P2	Handset Connector Card
SW5-P11	J21-P4	Magneto Call Switch Card
SW5-P12	J21-P6	Magneto Call Switch Card

Table 4: Alternative Connectors Versus FRC Connector Pins

RESTRICTED

5 PARTS LIST

5.1 Spare Part Kits

Part no.	Description
NTMR2012312	Spare Part Kit, Main Card
NTMR2012313	Spare Part Kit, Ringer Card
NTMR2012314	Spare Part Kit, System Switch Card
NTMR2012315	Spare Part Kit, Reed Switch Card
NTMR2012316	Spare Part Kit, Binding Post Card
NTMR2012317	Spare Part Kit, Volume Switch Card
NTMR2012318	Spare Part Kit, Magneto Call Switch Card
NTMR2012319	Spare Part Kit, Handset Connector Card
NTMR2012320	Spare Part Kit, Buzzer Assembly
NTMR2012321	Spare Part Kit, Battery Box Assembly
NTMR2012322	Spare Part Kit, Top Housing (Mechanical)
NTMR2012323	Spare Part Kit, Bottom Plate (Mechanical)
NTMR2012324	Spare Part Kit, Binding Posts
NTMR2012325	Spare Part Kit, Gasket Set
NTMR2012326	Spare Part Kit, Knobs and Handset Guards
NTMR2012327	Spare Part Kit, Rubber Parts
NTMR2012328	Spare Part Kit, Cable Harness Set
NTMR2012329	Spare Part Kit, Battery Box Cover Plate
NTMR2012330	Spare Part Kit, Screws and Washers

Table 5: Spare Part Kits

5.2 Accessories

Part no.	Description
AL60040	Handset
P73000090	Carrying bag

Table 6: Accessories

RESTRICTED

5.3 Parts Lists

5.3.1 Top Level, Parts List

DPAR002013-R1C		Document number: 13132-DPAR002013		Rev.: D	
AFT1001		Author: Ghanshyam Sharma		Date: 2021-02-24	
		Approved: K. Rønning			
REF. DES	ITEM ID	VER	DESCRIPTION	QTY	UOM
1	DPAR002013/1-R1	C	AFT1001 Assembly	1,00	EA
2	AL60040-R1	B	Handset	1,00	EA

Table 7: Top Level, Parts List

5.3.2 Assembly, Parts List

DPAR002013/1-R1C		Document number: 13132-DPAR002013/1		Rev.: C	
AFT1001 Assembly		Author: Ghanshyam Sharma		Date: 2021-05-20	
		Approved: K. Rønning			
REF. DES	ITEM ID	VER	DESCRIPTION	QTY	UOM
4	P46011138	-	AFT1001 Main Gasket	1,00	EA
5	P78000951		Screw M3x8, SS316, PH, TORX, Black	10,00	EA
6	P78000929	-	AFT1001 Top Housing	1,00	EA
7	P78001036	-	Handset Guard	2,00	EA
8	60787-R1	B	Keypad Assembly	1,00	EA
9	P78001037	-	System Switch Knob	1,00	EA
10	P78001038	-	Waterproof Cap for Magneto Call Switch	1,00	EA
11	P78001039	-	Volume Switch Knob	1,00	EA
12	P78000950		M3 Plain Washer, SS316	10,00	EA
13	P78000601	-	Screw M3x4, Brass, Slotted without head, Black	2,00	EA
14	P78000580	-	Screw M3x6, SS316, PH with washer, TORX	13,00	EA
15	P78000578	-	Screw M3x6, SS316, CSK, TORX, Black	6,00	EA
16	P46000503	-	Washer - Mica	21,00	EA
17	P46002072	-	O-Ring for Buzzer	1,00	EA
19	P46000586	-	System Switch Label	1,00	EA
20	P46000585	-	Battery Terminal Label	1,00	EA
21	P46000584	-	Line Terminal Label	1,00	EA
22	P46000587	-	Volume Switch Label	1,00	EA
23	P78000930	-	AFT1001 Bottom Plate	1,00	EA

DPAR002013/1-R1C		Document number: 13132-DPAR002013/1			Rev.: C
AFT1001 Assembly		Author: Ghanshyam Sharma			Date: 2021-05-20
		Approved: K. Rønning			
REF. DES	ITEM ID	VER	DESCRIPTION	QTY	UOM
24	P78000931	-	AFT1001 Battery Box Cover Plate	1,00	EA
26	P46011139		AFT1001 Battery Box Gasket	1,00	EA
27	P78000933	-	Thumb Screw, Black	4,00	EA
28	MVKR1022023	-	Dymax 3094 Adhesive	1.30	g
29	P78000969	-	Protective Vent with O-Ring	1,00	EA
31	P78000599	-	Screw M3x5, SS316, PH with washer, TORX	8,00	EA
32	P46000563	-	Feet	4,00	EA
33	P46011143	-	AFT1001 Instruction Label, French	1,00	EA
35	SVAR1032011	-	Polyester Label	1,00	EA
36	SVAR1032021/43		Metal Label	1,00	EA
38	P46000437	-	Loctite248	0.60	g
39	P78000991	-	M4 Plain Washer, SS316	4,00	EA
40	P46011111	-	Wire Mesh, SS316, 12mm	1,00	EA
41	P46000485		Loctite401	0.20	g
42	P46011140	-	Foam Tape, 35x1mm	133.00	mm
43	P78000982	-	Locating Ring, Magneto Call Switch	1,00	EA
44	P78000983	-	O-Ring, System Switch	1,00	EA
45	P78000984	-	Plain Washer, System Switch	1,00	EA
46	P78000985	-	Hex Nut, System Switch	1,00	EA
48	P78000987	-	Rubber Ring, Volume Switch	1,00	EA
49	P78000988	-	Lock Washer, Volume Switch	1,00	EA
50	P78000989	-	Hex Nut M6, Volume Switch	1.00	EA
51	P46000436	-	Dow Corning 3145	0.10	g
52	SVBR1292001/1	-	Label	2.00	EA
53	P78001035	-	Hex Nut, Magneto Call Switch	1.00	EA
54	MPPR1262012	-	Tape Double Sided, 25x1.1mm	5.00	mm
A4	1/FTL10001-R1	A	Main Card	1,00	EA
A5	3/FTL10001-R1	A	System Switch Card	1,00	EA
A6	4/FTL10001-R1	A	Reed Switch Card	1,00	EA
A7	5/FTL10001-R1	A	Binding Post Card	1,00	EA

DPAR002013/1-R1C		Document number: 13132-DPAR002013/1		Rev.: C	
AFT1001 Assembly		Author: Ghanshyam Sharma		Date: 2021-05-20	
		Approved: K. Rønning			
REF. DES	ITEM ID	VER	DESCRIPTION	QTY	UOM
A8	6/FTL10001-R1	A	Volume Switch Card	1,00	EA
A9	7/FTL10001-R1	A	Magneto Call Switch Card	1,00	EA
A10	8/FTL10001-R1	A	Handset Connector Card	1,00	EA
A11	60611	-	Buzzer Assembly	1,00	EA
A13	2/FTL10001-R1	A	Ringer Card	1,00	EA
A14	60622	-	Battery Box Assembly	1,00	EA
S1	P78000519	-	Black Cap for Binding Post	4,00	EA
S2	P78000518	-	Binding Post	4,00	EA
S3	P46000576	-	Washer Rubber M5	4,00	EA
S4	P46000573	-	Bush for Binding Post	4,00	EA
S5	P46000577	-	Washer Rubber M8	4,00	EA
S6	P46000575	-	Bush Fixture for Binding Post	4,00	EA
S7	P46000402	-	Washer Plain M5	12,00	EA
S8	P78000331	-	Nut M5	8,00	EA
W1	60833	-	External Battery Harness	1,00	EA
W2	60624	-	Line Terminal Harness	1,00	EA
W3	60625	-	Reed Switch Harness	1,00	EA
W4	60626	-	Magneto Call Switch Harness-1	1,00	EA
W5	60635	-	Magneto Call Switch Harness-2	1,00	EA
W6	60627	-	System Switch Harness-1	1,00	EA
W7	60636	-	System Switch Harness-2	1,00	EA
W8	60628	-	Volume Switch Harness	1,00	EA
W9	60629	-	Handset Connector Harness	1,00	EA
W10	60637	-	Battery Harness	1,00	EA

Table 8: Assembly, Parts List

6 Performance Reports

The objective of this feedback information system is to provide information on the indigenous equipment/systems regarding the general performance, the defects occurring in transportation, storage and use, to the Defense Quality Assurance Organization, R&D, Design Authority and Production Agency with a view to effective improvements in the same or similar equipment/systems.

Authority: 1. Army Order 59/78
 2. AIR HQ/43/695/1/STC-28-3-77
 3. Naval HQ

O Instructions

A. The seven performance Report Cards contained herein will be completed and dispatched as mentioned below :

Sl.N	Card N	Description	To be sent by	Period	Applicable to
1	I	Receipt/Installation Inspection by Consignee Depot/ on receipt from Establishment	Receiving Depot	After Receipt Inspection in Depot on receipt from Supplier	Army and Air Force only
2	II	Pre-Issue Inspection	Holding Depot	After Pre-Issue	Army and Air Force only
3	I	Receipt/Installation Inspection by consignee Depot / Establishment	Unit/Establishment receiving equipment for use	After Installation/ commissioning on receipt in Unit / Establishment	Army, Air Force and Navy only
4	III	Performance Report: Periodic (1 st)	“	6 months after commissioning	Navy only
		“	“	One year after commissioning	Army, Air Force only
5	III	Performance Report: Periodic (2 nd)	“	One year after commissioning	Navy only
		“	“	Two years after commissioning	Army, Air Force only
6	III	Performance Report: Periodic (3 rd)	“	One and a half years after commissioning	Navy only
		“	“	Three years after commissioning	Army, Air Force only
7	III	Performance Report: Periodic (4 th)	“	Three years after commissioning	Navy only
		“	“	Five years after commissioning	Army, Air Force only

Table 9: Performance Report Instructions

- B. Cards at Serial Nos.1 and 3 above are similar. However Card No. 1 at Serial No.1 will be completed at the receiving Depot and Card No. 1 and Serial No. 3 will be completed at the Unit/Establishment on receipt of equipment for installation /use.
- C. The Cards will be dispatched by Holding Depot/Establishment to the agencies mentioned in the relevant Service orders as indicated under Authority above (CQA(L),Bangalore / CQA(S), Secunderabad for Army equipment, ALISDA Bangalore/ DGAQA for Air Force equipment. NHQ (DEE) for Naval equipment, as applicable).
- D. Before dispatch of each Card, relevant information will be filled up in the counterfoil and initialed.

PERFORMANCE REPORT		Card No. I
(RECEIPT / INSTALLATION INSPECTION BY CONSIGNEE DEPOT / ESTABLISHMENT)		
A tick mark (✓) is to be placed in appropriate box		
COUNTER FOIL Defence Serial No. of Equipment : Card No. I Date Report Submitted Initial	A. To be filled by Manufacturer 1. Equipment (Service Designation) AICS 2. Cat Part No.(Pattern No./ Ref. No.) : 3. Indent No. and Date : 4. Date and Expiry of Warranty period * : 5. Defense Serial No. of equipment: 6. Manufacturer: * This entry is to enable prompt initiation of claim for free replacement in case of defect during warranty period. B. To be filled by Depot/Establishment receiving the Equipment 7. Date of receipt of equipment: 8. Receiving Depot/Unit : 9. Date of opening packages: 10. Mode of Transportation Train ☐ Goods ☐ Passenger ☐ Road ☐ Post ☐ Air ☐ 11. Condition of packages: Undamaged ☐ Damaged ☐ (Give details) 12. Condition of Delivery: Normal ☐ Open ☐ 13. Suggestions for improvement in packing/Packages if any	
	Date: Signature and designation	

C. To be filled in by Maintenance/Technical Authority

14 Condition of Equipment:

(a) Physical (Give details): Undamaged ☐ Damaged ☐

(b) Functional: Serviceable ☐ Repairable ☐ Unserviceable ☐

(c) Repairs/ adjustments carried out

Fault*	Assembly/Sub-Assembly	Component	Circuit Ref	Repair action taken
--------	-----------------------	-----------	-------------	---------------------

* Any deviations noted from specifications should also be listed here.

15. Additional remarks if any:

Address:

Quality Assurance Officer

Date :

(Signature and Designation)

Card No. II	Card No. II	PERFORMANCE REPORT
		Card No. II
(RECEIPT / INSTALLATION INSPECTION BY CONSIGNEE DEPOT / ESTABLISHMENT)		
A tick mark (✓) is to be placed in appropriate box		
A. To be filled by Manufacturer		
1. Equipment (Service Designation) :		
2. Cat Part No.(Pattern No./ Ref. No.) :		
3. Indent No. and Date :		
4. Date and Expiry of Warranty period * :		
5. Defense Serial No. of equipment :		
6. Manufacturer :		
* This entry is to enable prompt initiation of claim for free replacement in case of defect during warranty period.		
B. To be filled in by issuing Equipment Depot		
7. Issued to (Name of Unit) :		
8. Issuing Depot :		
9. Date of Issue :		
Date:		Signature and designation
C. To be filled in by Maintenance / Technical Authority:		
10. Remarks on Effect of Storage / Conditions of presentation (Give details of effects on finish, fungus growth etc.).		
11. Condition of Equipment:		
(a) Physical (Give details): Undamaged ☐ Damaged ☐		
(b) Functional: Serviceable ☐ Repairable ☐ Unserviceable ☐		
(c) Repairs/ adjustments carried out		

Fault*	Assembly/Sub-Assembly	Component	Circuit Ref	Repair action taken
--------	-----------------------	-----------	-------------	---------------------

* Any deviations noted from specifications should also be listed here.

12. Additional remarks if any:

Address:

Quality Assurance Officer

Date:

(Signature and Designation)

COUNTER FOIL	Card No. III	PERFORMANCE REPORT (PERIODIC)	Card No. III
Defence Serial No. of Equipment :	Date Report Submitted	Initial	Total Hours Run Upto Date:
A tick mark (✓) is to be placed in appropriate box			
A. To be filled by Manufacturer			
1. Equipment (Service Designation) :			
2. Cat Part No.(Pattern No./ Ref. No.) :			
3. Indent No. and Date :			
4. Date and Expiry of Warranty period * :			
5. Defense Serial No. of Equipment:			
6. Manufacturer :			
* This entry is to enable prompt initiation of claim for free replacement in case of defect during warranty period.			
B. To be filled in Holder of the Equipment			
7. Performance Report: Periodic		1st 2nd 3rd 4th	
8. Date of Receipt / Installation in Unit:			
9. Name of Unit:			
10. Employment of Equipment as:			
Static ☐ Mobile ☐ Manpack ☐ Track Ground Ship Air Borne			
11. Employment of Equipment:			
Protected Partially Protected Exposed			
12. Environment:			
Plains Mountains Sea Coast Highly Damp Desert Dusty Extreme Hot Extreme Cold			
13. Number of Hours Run ##			
a) Total upto date.....hours			
b) Till date of last report.....hours			
c) Total down time.....hours			
(Reason in brief)			
Date:.....		Signature and designation	
# Strike out except relevant one depending upon whether it is the 1 st , 2 nd , 3 rd or 4 th periodic report. ## 13(b) will be completed from the counter foil of previous ‘Periodic report’.			